

CMU 10-703: Deep Reinforcement Learning

Lecture 9: Off-Policy Actor-Critic and Maximum-Entropy RL

Max Simchowitz

Carnegie Mellon University

Lecture Roadmap

Off-policy actor-critic

Learn Q^π from replay data, then update the current policy through the learned critic.

DDPG and TD3

Deterministic policy gradients, twin critics, target smoothing, and delayed actor updates.

Maximum-entropy RL

Add future policy entropy to the objective and Bellman recursion.

Soft Actor-Critic

Combine TD3-style critics with a stochastic, entropy-regularized actor.

The central question is how replay-buffer data can train a stable, exploratory actor.

1.

Off-Policy Actor–Critic

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Two Performance-Difference Objectives

Form 1: PPO and AWR

$$\max_{\pi} \mathbb{E}_{s \sim d^{\pi}, a \sim \pi} [A^{\mu}(s, a)]$$

Form 1 evaluates candidate actions with the behavior policy's advantage.

Two Performance-Difference Objectives

Form 1: PPO and AWR

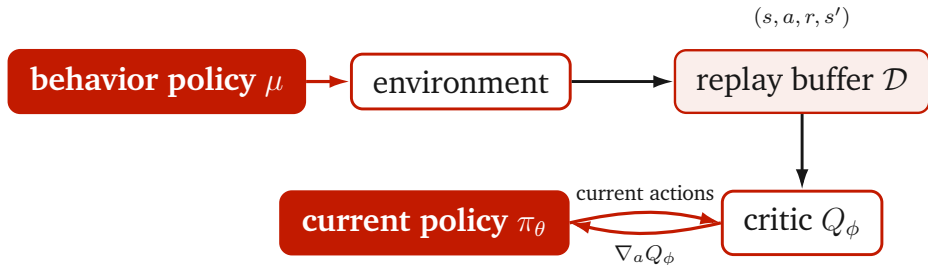
$$\max_{\pi} \mathbb{E}_{s \sim d^{\pi}, a \sim \pi} [A^{\mu}(s, a)]$$

Form 2: DDPG, TD3, and SAC

$$\max_{\pi} \mathbb{E}_{s \sim d^{\mu}, a \sim \pi} [A^{\pi}(s, a)]$$

Form 2 keeps replay-buffer states but evaluates the current policy.

Off-Policy Data Flow



Off-policy states

States and transitions come from d^μ , often across many earlier policies.

Current-policy actions

The actor supplies actions from π_θ , restoring a gradient path to θ .

Critic Bellman Equation

Critic target. The critic approximates the action value of the current policy π .

$$Q^\pi(s, a) = r(s, a) + \gamma \mathbb{E}_{\substack{s' \sim p(\cdot|s, a) \\ a' \sim \pi(\cdot|s')}} [Q^\pi(s', a')]$$

Observed

Replay supplies s, a, r, s' .

Resampled

Draw a' from the current policy.

Bootstrapped

Estimate the remaining return with a critic.

Critic Loss and Bellman Target

Squared Bellman error

$$L(\phi) = \mathbb{E}_{(s,a,r,s') \sim \mathcal{D}} \left[(Q_{\phi}(s,a) - y)^2 \right]$$

Fit the current critic to a one-step return estimate.

Critic Loss and Bellman Target

Squared Bellman error

$$L(\phi) = \mathbb{E}_{(s,a,r,s') \sim \mathcal{D}} \left[(Q_\phi(s,a) - y)^2 \right]$$

Target

$$y = r(s,a) + \gamma \mathbb{E}_{a' \sim \pi_\theta(\cdot|s')} \left[Q_{\bar{\phi}}^{\text{tgt}}(s',a') \right]$$

In practice, one sampled a' usually replaces the action expectation.

Target Networks

Online critic

$$Q_{\phi}(s, a)$$

Updated by gradient descent on every critic step.

Target critic

$$Q_{\bar{\phi}}^{\text{tgt}}(s', a')$$

Updated less often or by a slow Polyak average.



$$\bar{\phi} \leftarrow (1 - \tau)\bar{\phi} + \tau\phi$$

A delayed target prevents the regression target from moving as quickly as the fitted critic.

Zero Gradient for Replay Actions

Zero-gradient problem. A replay action $a \sim \mu$ is stored data, so it has no computational dependence on θ .

$$\nabla_{\theta} Q_{\phi}(s, a) = 0, \quad (s, a) \sim \mathcal{D}$$

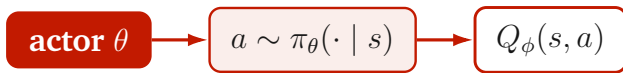


The critic can score replay actions, but that score cannot update the actor.

Actor Objective

Replace replay actions with current-policy actions

$$\max_{\theta} \mathbb{E}_{\substack{s \sim d^{\mu} \\ a \sim \pi_{\theta}(\cdot | s)}} [Q_{\phi}(s, a)]$$



The state remains off-policy; the action is regenerated by the policy being optimized.

Deterministic Policy Gradient

Deterministic actor. The policy maps each state directly to $a_\theta(s)$.

$$\nabla_\theta Q_\phi(s, a_\theta(s)) = \nabla_a Q_\phi(s, a)|_{a=a_\theta(s)} \nabla_\theta a_\theta(s)$$

Critic derivative

Which local change in action raises Q_ϕ ?

Actor derivative

How must θ change to produce that action change?

Stochastic Policy Reparameterization

Gaussian action sample

$$a_{\theta}(s) = f_{\theta}(s) + \Sigma_{\theta}(s)^{1/2}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

The random variable ϵ is independent of θ .

Stochastic Policy Reparameterization

Gaussian action sample

$$a_{\theta}(s) = f_{\theta}(s) + \Sigma_{\theta}(s)^{1/2}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

Reparameterization

Backpropagate through f_{θ} , Σ_{θ} , and the differentiable critic.

Score-function estimator

REINFORCE is available for discrete or non-reparameterizable policies.

Both estimators have the same gradient in expectation; SAC uses reparameterization.

2.

DDPG and TD3

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DDPG Components

Actor

A deterministic policy $a = \pi_{\theta}(s)$ for continuous actions.

Critic

$Q_{\phi}(s, a)$ estimates the current policy's action value.

Replay buffer

Reuses transitions collected by earlier noisy policies.

Source: Lillicrap et al., "Continuous Control with Deep Reinforcement Learning," 2015.

Target networks

Delayed actor and critic parameters stabilize bootstrapping.

Actor update

Backpropagates through $Q_{\phi}(s, \pi_{\theta}(s))$.

Exploration

Adds temporally correlated noise during rollout.

DDPG Critic and Actor Updates

Critic target

$$y = r + \gamma Q_{\bar{\phi}}(s', \pi_{\bar{\theta}}(s'))$$

Actor objective

$$\max_{\theta} \mathbb{E}_{s \sim \mathcal{D}} [Q_{\phi}(s, \pi_{\theta}(s))]$$

The target actor chooses the bootstrap action; the online actor receives the critic gradient.

DDPG Exploration Noise

Rollout policy. Exploration is added outside the deterministic actor during data collection.

$$a_t = \pi_{\theta}(s_t) + \epsilon_t$$

$$\epsilon_{t+1} = (1 - \kappa)\epsilon_t + \sigma\mathcal{N}(0, 1)$$

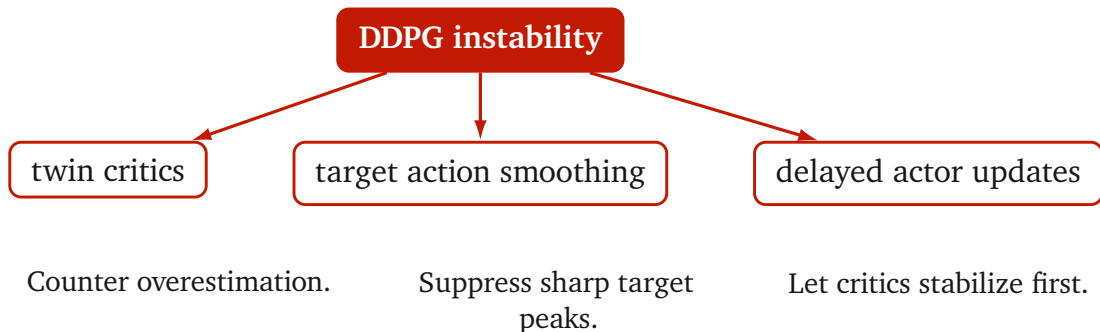
Temporal correlation

Ornstein–Uhlenbeck noise changes smoothly across time.

Mean reversion

κ pulls the exploration process back toward zero.

Three TD3 Improvements



Source: Fujimoto, van Hoof, and Meger, "Addressing Function Approximation Error in Actor-Critic Methods," ICML 2018.

Clipped Double Q-Learning

Twin-critic target

$$y(s, a) = r(s, a) + \gamma \min_{i \in \{1, 2\}} Q_{\phi_i}^{\text{tgt}}(s', a')$$

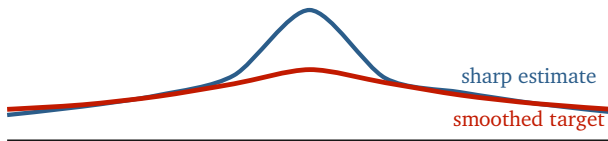
$$Q_{\phi_1}(s, a) \quad \longrightarrow \quad \min(Q_1, Q_2) \quad \longleftarrow \quad Q_{\phi_2}(s, a)$$

Because the actor maximizes the critic, a conservative target limits compounding overestimation.

Target Action Smoothing

Noisy target action

$$\tilde{a}' = \pi_{\theta}(s') + \varepsilon, \quad \varepsilon \sim \text{clip}(\mathcal{N}(0, \sigma^2), -c, c)$$



The target averages over a small action neighborhood, limiting exploitation of narrow critic errors.

Delayed Policy Updates

Every step

update Q_{ϕ_1}, Q_{ϕ_2}



Every d steps

update actor and targets

$d = 2 \implies$ two critic steps per actor step

The actor waits for a better critic before following its action gradient.

TD3 Training Sketch

1. Collect $a = \pi_{\theta}(s) + \varepsilon$ and store (s, a, r, s') in $\mathcal{D}$.

Data collection remains deterministic-policy exploration with additive noise.

TD3 Training Sketch

1. Collect $a = \pi_{\theta}(s) + \varepsilon$ and store (s, a, r, s') in $\mathcal{D}$.
2. Form a smoothed next action and clipped double-Q target.

$$y = r + \gamma \min_{i=1,2} Q_{\bar{\phi}_i}(s', \pi_{\bar{\theta}}(s') + \varepsilon')$$

The target contains both twin critics and target action smoothing.

TD3 Training Sketch

1. Collect $a = \pi_{\theta}(s) + \varepsilon$ and store (s, a, r, s') in $\mathcal{D}$.
2. Form a smoothed next action and clipped double-Q target.

$$y = r + \gamma \min_{i=1,2} Q_{\bar{\phi}_i}(s', \pi_{\bar{\theta}}(s') + \varepsilon')$$

3. Minimize $(Q_{\phi_i}(s, a) - y)^2$ for both critics.

Both critics learn from the same conservative target.

TD3 Training Sketch

1. Collect $a = \pi_\theta(s) + \varepsilon$ and store (s, a, r, s') in $\mathcal{D}$.
2. Form a smoothed next action and clipped double-Q target.

$$y = r + \gamma \min_{i=1,2} Q_{\bar{\phi}_i}(s', \pi_{\bar{\theta}}(s')) + \varepsilon'$$

3. Minimize $(Q_{\phi_i}(s, a) - y)^2$ for both critics.
4. Every d steps, maximize $Q_{\phi_1}(s, \pi_\theta(s))$ and soft-update all targets.

The actor and all targets move on the delayed schedule.

3.

Maximum-Entropy Reinforcement Learning

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Standard and Maximum-Entropy Objectives

Standard RL

$$\pi_{\text{RL}}^* = \arg \max_{\pi} \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right]$$

Maximum-entropy RL

$$\pi_{\text{MaxEnt}}^* = \arg \max_{\pi} \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t (r(s_t, a_t) + \alpha \mathcal{H}(\pi(\cdot \mid s_t))) \right]$$

Entropy and Temperature

Shannon entropy. The expected negative log-probability of an action under the policy.

$$\mathcal{H}(\pi(\cdot | s)) = \mathbb{E}_{a \sim \pi(\cdot | s)} [-\log \pi(a | s)]$$

Low temperature α

Reward dominates, favoring concentrated policies.

High temperature α

Entropy receives more weight, preserving action diversity.

Policy-Dependent Modified Reward

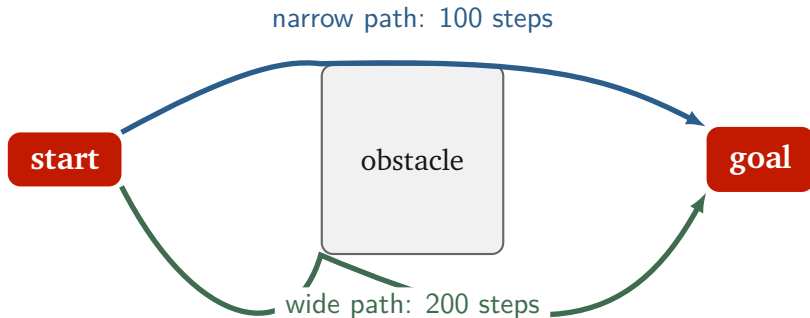
Equivalent one-step reward

$$\tilde{r}_\alpha(s, a) = r(s, a) - \alpha \log \pi(a \mid s)$$



Entropy enters every future backup, so the policy values states that permit diverse continuation actions.

Navigation with Narrow and Wide Routes



Standard RL

Selects the shortest route and can become brittle to a shifted obstacle.

Maximum-entropy RL

Maintains multiple trajectory variations and tolerates environmental change.

Properties of Maximum-Entropy Policies

Stochastic optimum

Unlike an unregularized finite MDP, the optimal policy generally spreads mass across actions.

Behavioral diversity

Multiple near-optimal strategies remain available.

Robustness

Diverse policies can tolerate reward perturbations and noise.

Optimization

Entropy smooths the objective and strengthens concavity.

Source: Robustness interpretation: Eysenbach and Levine (2022), as cited in the lecture notes.

Entropy Regularization Versus Entropy Backup

Add exploration noise afterward

optimize original reward



add current-step noise

The underlying optimal policy remains the standard RL solution.

Back up entropy

optimize modified reward



seek high-entropy futures

The Bellman recursion changes, producing a different optimal policy.

4.

Soft Bellman Equations and Policy Iteration

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Soft Action and State Values

Soft action value

$$Q_{\text{soft}}^{\pi}(s, a) = r(s, a) + \gamma \mathbb{E}_{\substack{s' \sim p(\cdot | s, a) \\ a' \sim \pi(\cdot | s')}} [Q_{\text{soft}}^{\pi}(s', a') - \alpha \log \pi(a' | s')]$$

Soft state value

$$V_{\text{soft}}^{\pi}(s) = \mathbb{E}_{a \sim \pi(\cdot | s)} [Q_{\text{soft}}^{\pi}(s, a) - \alpha \log \pi(a | s)]$$

Soft Policy Evaluation

Evaluation step. Hold π fixed and repeatedly apply the soft Bellman backup.

$$Q(s, a) \leftarrow r(s, a) + \gamma \mathbb{E}_{a' \sim \pi(\cdot | s')} [Q_{\text{target}}(s', a') - \alpha \log \pi(a' | s')]$$

The target values both future reward and the entropy retained by the policy.

Soft Policy Improvement Objective

Improvement step. At each state, choose a policy that balances high Q^π with high entropy.

$$\pi_{\text{new}} = \arg \max_{\pi} \mathbb{E}_{a \sim \pi(\cdot | s)} [Q^\pi(s, a) - \alpha \log \pi(a | s)]$$

Value term

Raises probability on actions with larger $Q^\pi(s, a)$.

Entropy term

Prevents immediate collapse to a point mass.

Boltzmann Policy Improvement

Closed-form improved policy

$$\pi_{\text{new}}(a \mid s) = \frac{\exp(Q^{\pi}(s, a)/\alpha)}{Z(s)}$$

The partition function $Z(s)$ normalizes the action distribution.

Boltzmann Policy Improvement

Closed-form improved policy

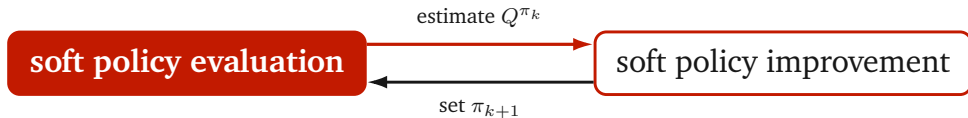
$$\pi_{\text{new}}(a \mid s) = \frac{\exp(Q^\pi(s, a)/\alpha)}{Z(s)}$$

Equivalent KL projection

$$\min_{\pi} D_{\text{KL}} \left(\pi(\cdot \mid s) \parallel \frac{\exp(Q^\pi(s, \cdot)/\alpha)}{Z(s)} \right)$$

Policy improvement projects the policy toward the exponentiated soft value.

Soft Policy Iteration



Monotonic improvement

$$Q^{\pi_{k+1}}(s, a) \geq Q^{\pi_k}(s, a) \quad \text{for all } (s, a)$$

Repeated evaluation and improvement converge to the maximum-entropy optimum.

5.

Soft Actor-Critic

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SAC Components

Twin critics

Clipped double-Q targets reduce overestimation.

Stochastic actor

A reparameterized policy produces exploration directly.

Temperature tuning

α adapts online toward a target entropy.

Off-policy machinery

Replay and target networks follow the TD3 pattern.

SAC combines TD3-style value learning with maximum-entropy policy improvement.

Source: Haarnoja et al., "Soft Actor-Critic," ICML 2018.

SAC Critic Update

Entropy-augmented clipped double-Q target

$$y = r(s, a) + \gamma \mathbb{E}_{a' \sim \pi_\theta(\cdot | s')} \left[\min_{i=1,2} Q_{\bar{\phi}_i}(s', a') - \alpha \log \pi_\theta(a' | s') \right]$$

Critic losses

$$\min_{\phi_i} \mathbb{E}_{(s,a,r,s') \sim \mathcal{D}} \left[(Q_{\phi_i}(s, a) - y)^2 \right], \quad i \in \{1, 2\}$$

SAC Actor Update

Entropy-regularized actor objective

$$\max_{\theta} \mathbb{E}_{\substack{s \sim \mathcal{D} \\ a \sim \pi_{\theta}(\cdot | s)}} [Q_{\phi}(s, a) - \alpha \log \pi_{\theta}(a | s)]$$

$$a = f_{\theta}(s) + \Sigma_{\theta}(s)^{1/2} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

The reparameterized action carries gradients from both Q_{ϕ} and $\log \pi_{\theta}$.

Automatic Temperature Tuning



Entropy below target

Increase the pressure for broader action distributions.

Entropy above target

Allow reward maximization to receive more relative weight.

Adaptive temperature removes a fixed reward–entropy tradeoff from manual tuning.

Soft Actor-Critic Algorithm

1. Initialize replay buffer $\mathcal{D}$, stochastic policy π_θ , twin critics, and target critics.

SAC begins with the same replay and target-network infrastructure as TD3.

Soft Actor-Critic Algorithm

1. Initialize replay buffer $\mathcal{D}$, stochastic policy π_θ , twin critics, and target critics.
2. Sample $a_t \sim \pi_\theta(\cdot \mid s_t)$, execute it, and store (s_t, a_t, r_t, s_{t+1}) .

Policy sampling supplies exploration without separate rollout noise.

Soft Actor-Critic Algorithm

1. Initialize replay buffer $\mathcal{D}$, stochastic policy π_θ , twin critics, and target critics.
2. Sample $a_t \sim \pi_\theta(\cdot \mid s_t)$, execute it, and store (s_t, a_t, r_t, s_{t+1}) .
3. Sample a minibatch and update both critics with the entropy-augmented clipped double-Q target.

The critic target includes both conservatism and future entropy.

Soft Actor-Critic Algorithm

1. Initialize replay buffer $\mathcal{D}$, stochastic policy π_θ , twin critics, and target critics.
2. Sample $a_t \sim \pi_\theta(\cdot \mid s_t)$, execute it, and store (s_t, a_t, r_t, s_{t+1}) .
3. Sample a minibatch and update both critics with the entropy-augmented clipped double-Q target.
4. Update the actor through reparameterized samples and update α toward the target entropy.

Actor and temperature updates implement soft policy improvement.

Soft Actor-Critic Algorithm

1. Initialize replay buffer $\mathcal{D}$, stochastic policy π_θ , twin critics, and target critics.
2. Sample $a_t \sim \pi_\theta(\cdot \mid s_t)$, execute it, and store (s_t, a_t, r_t, s_{t+1}) .
3. Sample a minibatch and update both critics with the entropy-augmented clipped double-Q target.
4. Update the actor through reparameterized samples and update α toward the target entropy.
5. Soft-update target critics.

$$\bar{\phi}_i \leftarrow (1 - \tau)\bar{\phi}_i + \tau\phi_i, \quad i \in \{1, 2\}$$

Slow targets complete the off-policy update loop.

Training and Evaluation Policies

Training

$$a \sim \pi_{\theta}(\cdot \mid s)$$

Sampling preserves the entropy objective and gathers diverse data.

Evaluation

$$a = f_{\theta}(s)$$

The deterministic mean removes sampling variance at test time.

6.

Applications and Comparison

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Continuous-Control Benchmarks

MuJoCo tasks

HalfCheetah · Hopper · Walker2d · Ant · Humanoid

Sample efficiency

SAC learns with fewer environment interactions than DDPG and TD3.

Variance

Performance varies less across random seeds.

Hard tasks

Asymptotic gains are strongest on tasks such as Humanoid.

Source: Haarnoja et al., "Soft Actor-Critic," ICML 2018, as summarized in the lecture notes.

Robotics Applications



Real-world learning

The lecture notes report quadruped locomotion learned in approximately 20 minutes.

Rich action spaces

Stochastic policies support exploration in continuous, high-dimensional manipulation tasks.

Maximum-Entropy Use Cases

1. **Multiple valid behaviors:** preserve several successful strategies instead of collapsing to one.
2. **Robustness:** tolerate reward perturbations, noise, and environmental changes.
3. **Difficult optimization:** use entropy to smooth the objective and accelerate convergence.
4. **Transfer and adaptation:** reuse diverse behaviors for new tasks with limited additional data.

Maximum entropy is most useful when diversity has value beyond immediate random exploration.

DDPG, TD3, and SAC

Feature	DDPG	TD3	SAC
Policy type	Deterministic	Deterministic	Stochastic
Critics	1	2 (twin)	2 (twin)
Target smoothing	No	Yes	No
Delayed updates	No	Yes	No
Entropy bonus	No	No	Yes, auto-tuned α
Exploration	OU noise	Gaussian noise	Policy sampling

DDPG

One critic and a deterministic actor.

TD3

Three stability corrections for DDPG.

SAC

Twin critics plus soft policy improvement.

Lecture Summary

1. Off-policy actor–critic uses replay states, a Bellman-trained Q^π , and actions from the current policy.
2. DDPG differentiates a deterministic actor through its critic.
3. TD3 adds twin critics, target action smoothing, and delayed actor updates.
4. Maximum-entropy RL backs future policy entropy into soft values and policy improvement.
5. SAC combines twin soft critics, a reparameterized stochastic actor, and adaptive temperature.

Replay provides data efficiency; conservative critics and entropy make the actor update more stable and diverse.